Supplementary Information for Risk-Aware Template-Switching Control under Partial Quantum Observability

Supplementary Contents

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S1 Global constraint and clock-conditioned dynamics

Sections S1–S9 preserve the submitted relational formulation and its historical diagnostics for audit. They are not performance ingredients of R3 or R4, whose observation model and decision comparisons are given in Sections S10 and S11. The historical total Hilbert space is written as

$$\mathcal{H}_{\mathrm{tot}} = \mathcal{H}_C \otimes \mathcal{H}_S, \tag{S1.1}$$

and physical states are taken to satisfy the global constraint

$$\hat{H}_{\mathrm{tot}}|\Psi\rangle = 0. \tag{S1.2}$$

For a separable Hamiltonian

$$\hat{H}_{\mathrm{tot}} = \hat{H}_C \otimes I_S + I_C \otimes \hat{H}_S, \tag{S1.3}$$

a non-trivial product-basis solution requires matching clock and system energies,

$$\mathrm{Spec}(\hat{H}_C) \cap \mathrm{Spec}(-\hat{H}_S) \neq \emptyset. \tag{S1.4}$$

Indeed, in the product eigenbasis of $\hat{H}_C$ and $\hat{H}_S$, each component has total energy $E_k^C + E_j^S$. A non-zero vector in the kernel can therefore have support only on pairs satisfying $E_k^C = -E_j^S$. This is the finite-dimensional version of the compatibility condition used in the simulations. It is not used in the main text as an additional experimental assumption; it records why the clock-system construction is internally consistent.

This condition is used as a consistency requirement for constructing finite clock-system simulations; the main paper does not rely on a stronger claim that the finite simulation realizes a complete theory of quantum time.

Clock labels are generated by the clock orbit

$$|\tau\rangle_C = e^{-i\hat{H}_C\tau}|0\rangle_C. \tag{S1.5}$$

Conditioning the global state on $|\tau\rangle_C$ gives

$$\rho_S(\tau) = \frac{\mathrm{Tr}_C[(|\tau\rangle\langle\tau|_C \otimes I_S)|\Psi\rangle\langle\Psi|]}{\mathrm{Tr}[(|\tau\rangle\langle\tau|_C \otimes I_S)|\Psi\rangle\langle\Psi|]}. \tag{S1.6}$$

All trajectories in the manuscript are ordered families over these internal clock labels. They should not be read as externally time-parametrized histories. The finite clock representation also imposes resolution limits. Neighboring clock labels must be distinguishable enough to support separate conditional slices, but the clock orbit remains finite and can recur. These effects are handled by using a fixed ordered set of clock labels and by calibrating clock-consistency thresholds from reference trajectories rather than asserting an ideal continuum clock.

The submitted implementation, retained only as an audit dataset, fixed the clock object rather than tuning it per regime. The clock Hilbert space is one qubit, the initial clock state is $|+\rangle_C$, and

$$H_C = 0.8X_C. \tag{S1.7}$$

The sampled internal labels are

$$\tau_r \in \{-2.0, -1.5, -1.0, -0.5, 0, 0.5, 1.0, 1.5, 2.0\}, \qquad m = 9. \tag{S1.8}$$

The same clock Hamiltonian, clock grid, and ordering convention are used for the syndrome-first candidate, the relational candidate, the reference bank, and all policy comparisons.

S2 Spectral deformation and detectability

At each clock slice, the system state $\rho_S(\tau)$ is mapped to a weighted relational graph with adjacency matrix $W^{(\tau)}$. The graph Laplacian is

$$L(\tau) = D^{(\tau)} - W^{(\tau)}, \qquad D_{ii}^{(\tau)} = \sum_{j\neq i} W_{ij}^{(\tau)}. \tag{S2.1}$$

The graph functional is fixed by the code family in the reported experiments. For the phase-flip repetition code, the edge weight is the absolute two-qubit coherence

$$W_{ij}^{(\tau)} = \sum_{a\neq b} \left| \left(\rho_{ij}^{(\tau)}\right)_{ab} \right|. \tag{S2.2}$$

For the bit-flip repetition code, the edge weight is the two-qubit mutual information

$$W_{ij}^{(\tau)} = S(\rho_i^{(\tau)}) + S(\rho_j^{(\tau)}) - S(\rho_{ij}^{(\tau)}), \tag{S2.3}$$

where $S(\rho) = -\mathrm{Tr}(\rho \log \rho)$. Other relational functionals are available in the codebase for diagnostics, but they are not mixed into the reported C_2 versus C_3^{R} comparisons. The code-dependent choice is fixed before policy evaluation and is not selected per regime or per noise family. We use the Laplacian spectrum rather than the density-matrix spectrum directly because the decision variable is the collective connectivity pattern induced by code-specific two-qubit relations, not the global eigenvalue distribution of the state itself. Under a node relabeling the Laplacian changes only by permutation similarity, so its spectrum summarizes relational structure without depending on the arbitrary ordering of the three data qubits. Noise and partial observation induce perturbations

$$L(\tau) \mapsto L(\tau) + \Delta L(\tau). \tag{S2.4}$$

If $\lambda_k(\tau)$ and $\tilde{\lambda}_k(\tau)$ denote the ordered spectra before and after perturbation, Weyl-type perturbation control gives

$$|\tilde{\lambda}_k(\tau) - \lambda_k(\tau)| \leq \|\Delta L(\tau)\|_2. \tag{S2.5}$$

This bound explains why the Laplacian spectrum is an appropriate diagnostic object: perturbations in the relational graph create controlled spectral deformations, and the trajectory-level distance accumulates those deformations over clock labels. The experiments do not require every microscopic error to be separately identifiable. They require only that the deformation relevant to the recovery decision be large enough relative to calibrated background variation.

Consequently, non-trivial spectral deformations are detectable in principle when they are resolved by the spectral estimator and exceed the calibrated background variation. In the numerical implementation, this principle is used conservatively: detection and recovery are judged by empirical trajectory distances, calibrated admissibility penalties, and lower-tail recovery metrics rather than by a claim of universal noise detectability.

S3 Relational trajectory-space formalism

The clock-conditioned spectra define a spectral family

$$\Gamma_\Psi(\tau) = \Lambda(\tau) = (\lambda_1(\tau), \ldots, \lambda_n(\tau)). \tag{S3.1}$$

For sampled clock labels $\{\tau_r\}_{r=1}^m$, the computational distance between two spectral families is

$$\begin{aligned} d_{\mathcal{M}}^{ab} &\equiv d_{\mathcal{M}}(\Gamma^{(a)}, \Gamma^{(b)}), \\ d_{\mathcal{M}}^{ab} &= \left(\sum_{r=1}^{m} w_r \left\| \Lambda^{(a)}(\tau_r) - \Lambda^{(b)}(\tau_r) \right\|_2^2 \right)^{1/2}. \end{aligned} \tag{S3.2}$$

We use the term relational spectral manifold only as a geometric shorthand for this constrained metric trajectory space. The reported experiments use the sampled spectra and the metric $d_{\mathcal{M}}$ directly. Differential-geometric terms such as tangent vectors, geodesic curves, and Christoffel symbols are therefore interpretive language, not assumptions required by the recovery algorithm or by the policy validation. Boundary points, discrete reference banks, and piecewise admissible regions are allowed in the computational formulation.

This is also why the construction is not an ordinary temporal smoother. A standard smoother regularizes an externally ordered sequence. Here the ordering comes from an internal clock-conditioned family, and the same candidate must satisfy Laplacian validity, clock consistency, spectral smoothness, and reference-anchor compatibility. The role of this section is therefore claim calibration: it explains the metric object being audited without asserting a foundational smooth-manifold theory.

S4 Admissibility penalties and recovery algorithm

The admissible set is represented as an intersection of constraint families,

$$\mathcal{A} = \mathcal{A}_{\mathrm{lap}} \cap \mathcal{A}_{\mathrm{rel}} \cap \mathcal{A}_{\mathrm{ref}}. \tag{S4.1}$$

The implementation audits membership through four penalties. The Laplacian penalty checks whether each slice remains a valid weighted-graph Laplacian:

$$\Phi_{\mathrm{lap}}(\Gamma) = \sum_{r=1}^{m} \Big(\|L_r - L_r^\top\|_F^2 + \|L_r \mathbf{1}\|_2^2 + \sum_k [-\lambda_k(L_r)]_+^2 + \sum_{i \neq j} [L_{r,ij}]_+^2 + \left\| \mathrm{diag}(L_r) + \sum_{j \neq i} L_{r,ij} \right\|_2^2 \Big), \tag{S4.2}$$

where $[x]_+ = \max(x, 0)$. The terms enforce symmetry, zero row sum, positive-semidefinite spectrum, non-positive off-diagonal entries, and diagonal consistency.

The smoothness penalty measures slice-to-slice spectral jumps:

$$\Phi_{\mathrm{smooth}}(\Gamma) = \sum_{r=1}^{m-1} \frac{\|\Lambda(\tau_{r+1}) - \Lambda(\tau_r)\|_2^2}{(\tau_{r+1} - \tau_r)^2}. \tag{S4.3}$$

The clock-consistency penalty makes a large spectral jump more severe when the neighboring clock states are not well separated:

$$\Phi_{\mathrm{clock}}(\Gamma) = \sum_{r=1}^{m-1} \frac{\|\Lambda(\tau_{r+1}) - \Lambda(\tau_r)\|_2}{1 - |\langle \tau_r | \tau_{r+1} \rangle|^2 + \epsilon}. \tag{S4.4}$$

The reference-bank penalty is

$$\Phi_{\mathrm{ref}}(\Gamma) = \min_{\Gamma' \in \mathcal{R}} d_{\mathcal{M}}(\Gamma, \Gamma')^2. \tag{S4.5}$$

Thresholds are calibrated from reference trajectories as

$$\theta_k = \mu_k + \kappa \sigma_k, \qquad \kappa = 1.5, \tag{S4.6}$$

with leave-one-out evaluation for the reference penalty so that a reference trajectory does not calibrate its own distance to zero. For a candidate Γ_X, the normalized violation used by the robust policy is

$$v_X = \max_k \max \left\{ 0, \frac{\Phi_k(\Gamma_X) - \theta_k}{\theta_k + \epsilon} \right\}. \tag{S4.7}$$

The use of a maximum rather than an average is important for the robust policy. Average violation can hide a single severe constraint failure, while the maximum records the worst active excursion beyond threshold. This is the value denoted by v_A and v_B in the C_3^{R} gate.

The reference bank contains three clean trajectories for each encoded code: the logical reference state and two $\pm 20^\circ$ logical perturbations. These reference trajectories use the same clock grid, clock Hamiltonian, and graph functional as the observed row, but the noise channel is removed. The reference anchor is selected only by trajectory distance:

$$\Gamma_{\mathrm{ref,anchor}} = \arg\min_{\Gamma' \in \mathcal{R}} d_{\mathcal{M}}(\Gamma', \Gamma_{\mathrm{obs}})^2. \tag{S4.8}$$

No clean logical outcome, candidate fidelity, or oracle row label is used during anchor selection. This separation is important because the reference anchor is part of candidate construction, whereas fidelity and oracle quantities are held out for evaluation. The three-anchor bank is intentionally minimal. It is used to test whether a small auditable reference family can support recovery-side control, not to approximate a dense clean-state manifold. The recovery objective is

$$\begin{aligned} J_{\mathrm{rec}}(\Gamma) =& \alpha d_{\mathcal{M}}(\Gamma, \Gamma_{\mathrm{obs}})^2 + \beta d_{\mathcal{M}}(\Gamma, \Gamma_{\mathrm{ref,anchor}})^2 \\ &+ \lambda_1 \Phi_{\mathrm{lap}}(\Gamma) + \lambda_2 \Phi_{\mathrm{smooth}}(\Gamma) + \lambda_3 \Phi_{\mathrm{clock}}(\Gamma). \end{aligned} \tag{S4.9}$$

The main experiments use the discrete admissible projection as the default operator. A continuous convex refinement over local admissible combinations can be used as a diagnostic extension, but it is not required for the main reported policy results. The reason is operational: the first-stage operator selects from a finite, predeclared candidate set, so each accepted recovery can be traced to a reference anchor, a recovery objective value, and a vector of admissibility violations. This keeps the candidate construction separate from the downstream fidelity evaluation and prevents the reported policy result from depending on hidden continuous optimization flexibility. The discrete projection stage evaluates a finite set of reference-compatible candidate trajectories and selects the one minimizing the recovery objective while respecting admissibility. If $\mathcal{C}_{\mathrm{ref}}$ is the finite reference-compatible candidate set, the first-stage recovery is

$$\Gamma_1 = \arg\min_{\Gamma \in \mathcal{C}_{\mathrm{ref}}} J_{\mathrm{rec}}(\Gamma) \quad \text{s.t.} \quad \Phi_k(\Gamma) \le \theta_k \ \ \forall k. \tag{S4.10}$$

The optional refinement stage considers convex combinations of nearby admissible candidates. For a local neighborhood $\mathcal{N}_K(\Gamma_1) = \{\Gamma_j\}_{j=1}^K$, the feasible hull is

$$\mathcal{H}_K(\Gamma_1) = \left\{ \sum_{j=1}^K \alpha_j \Gamma_j : \alpha_j \ge 0,\ \sum_{j=1}^K \alpha_j = 1 \right\}. \tag{S4.11}$$

The diagnostic second-stage candidate is then

$$\Gamma_2 = \arg\min_{\Gamma \in \mathcal{H}_K(\Gamma_1)} J_{\mathrm{rec}}(\Gamma) \quad \text{s.t.} \quad \Phi_k(\Gamma) \le \theta_k \ \ \forall k. \tag{S4.12}$$

The reported C3R policy comparisons use the conservative application rule

$$\Gamma_{\mathrm{rec}} = \Gamma_1, \tag{S4.13}$$

while Γ_2 is retained only for sensitivity diagnostics. This two-stage design keeps the paper-facing operator auditable while leaving room for continuous diagnostics. In particular, it avoids post-hoc candidate shaping, preserves row-level interpretability, and makes the intervention metrics attributable to the policy gate rather than to an additional refinement step. Stage 2 is therefore reported as a diagnostic stress test of the recovery geometry, not as a second source of final-policy performance.

S5 Encoded-QEC simulation protocol

The encoded-QEC testbed uses three-qubit repetition codes in bit-flip and phase-flip configurations. Noise families include bit-flip, phase-flip, depolarizing, dephasing, and mixed-Pauli channels. Each simulation row constructs two candidates:

$$A = \text{syndrome-first decoded recovery}, \qquad B = \text{relational admissible recovery}. \tag{S5.1}$$

Candidate A is the conventional decoder-facing baseline. Candidate B is the output of admissible spectral recovery. The policy layer never changes the candidate construction; it only decides which candidate to trust on each row. This design isolates policy behavior. If the candidates themselves changed across partial, noisy, or ambiguous syndrome protocols, it would be difficult to tell whether an observed improvement came from better recovery construction or from a better decision rule. Here the decision layer is tested against a fixed pair of candidate recoveries, while syndrome observability affects the policy evidence and uncertainty diagnostics.

Syndrome observability is controlled by an observation ratio, syndrome-bit noise, ambiguity, measurement error, reset error, and measured information loss. The uncertainty proxy used by the robust gate is

$$u_{\text{raw}} = (1 - r_{\text{obs}}) + p_{\text{noise}} + a_{\text{amb}} + p_{\text{meas}} + p_{\text{reset}} + \ell_{\text{syn}}, \qquad u_{\text{syn}} = \text{clip}_{[0,1]}(u_{\text{raw}}). \tag{S5.2}$$

Figures S1 and S2 show the explicit circuit-level implementation of the testbed. The encoder circuits draw the three-qubit repetition-code conventions: bit-flip protection uses two CNOTs from the data qubit q_0 to q_1 and q_2, while phase-flip protection appends Hadamards on all three qubits to map the bit-flip stabilizers into the X basis. The recovery circuits draw the corresponding syndrome-extraction circuits with two ancillas a_0, a_1 measured into classical bits s_0, s_1 and feed-forward X corrections gated on the syndrome values $\{0\text{x}1, 0\text{x}2, 0\text{x}3\}$. The clock-conditioned data state $\rho_{\text{data}}(\tau)$ is injected on $\{q_0, q_1, q_2\}$ at each clock slice; the relational graph functional above is computed from this injected state before any syndrome operation. The stress protocols used in the main analysis are clean syndrome, partial syndrome, noisy syndrome, partial-plus-noisy syndrome, and ambiguity with measurement/reset errors. The clean map identifies operating zones when direct syndrome information is reliable. The partial and partial-plus-noisy maps test missing evidence and combined degradation. The noisy-only map tests random syndrome-bit corruption without reducing observation ratio. The ambiguity-plus-measurement map tests whether uncertainty due to ambiguous or physically corrupted syndrome evidence produces the same risk-control behavior.

S6 Policy maps and operating-zone analysis

The four policies compared in the manuscript are deterministic maps from the same two candidates to one selected recovery. C_1 is the conservative rule-based hybrid. C_2 is the balanced proposal policy with score

$$S_X^{(2)} = \lambda_s s_X - \lambda_t d_X - \lambda_i (1 - a_X) + \lambda_o \frac{1}{1 + J_X}. \tag{S6.1}$$

C_3 is a hard-admissibility ablation: it selects the uniquely admissible candidate when exactly one candidate is admissible, and otherwise uses a safety-weighted score. The clean-regime result that C_3 does not form an independent operating zone is interpreted as an ablation result, not as the central claim of the paper. The reason for retaining C_3 is methodological. It tests the simpler hypothesis that strict admissibility alone is enough to create a useful safety-first region. The negative operating-zone result indicates that this Boolean criterion is too coarse once inadmissibility has already been priced into the balanced score used by C_2. This motivates the one-sided veto design of C_3^{R}.

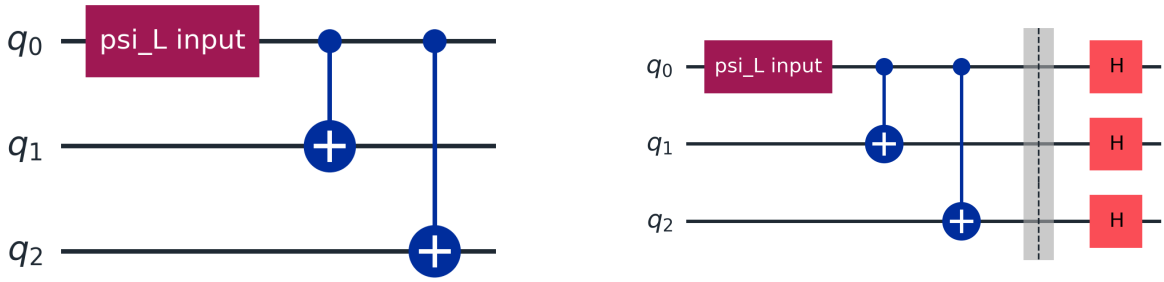


Figure S1: Encoder schematics for the encoded-QEC testbed. The left panel shows the bit-flip repetition-code encoder and the right panel shows the phase-flip encoder obtained by adding Hadamards on all three data qubits. The label ψ_L denotes the logical input state used to initialize the repetition-code block.

The revised policy C_3^{R} treats C_2 as a proposal and applies a one-sided veto:

$$C_3^{\mathrm{R}} = B \iff [C_2 = B] \wedge [\Delta S \geq \delta_S] \wedge [v_A \geq \delta_A] \wedge [B \text{ admissible}, v_B \leq \delta_B] \wedge [u_{\mathrm{syn}} \leq u_{\max}]. \quad \text{(S6.2)}$$

The main operating point is

$$\delta_S = 0, \qquad \delta_A = 0, \qquad \delta_B = 0, \qquad u_{\max} = 0.99. \quad \text{(S6.3)}$$

Thus

$$C_3^{\mathrm{R}} = B \Rightarrow C_2 = B, \quad \text{(S6.4)}$$

so the revised controller can block proposed relational switches but cannot create new ones. The useful diagnostics for C_3^{R} are therefore conditional on C_2 proposing B. Unconditional switch rates can be misleading because most rows may already be preserved by C_2. The intervention analysis in the main text and result tables consequently report block rate given $C_2 = B$, harmful-block precision, harmful-switch recall, beneficial-switch block rate, and net intervention effect. Beneficial-switch retention is the complementary quantity.

Table S1: Clean-regime preferred-policy counts over the 240-cell operating map. Each cell is averaged over ten random seeds.

Policy	Cells	Share	Interpretation
C_1	197	0.8208	Default conservative hybrid region.
C_2	43	0.1792	Structured correction region; all cells are phase-flip code rows.
C_3	0	0.0000	No independent strict-admissibility zone.
C_3^{R}	0	0.0000	No clean-regime veto trigger at the default gate.

Table S2: Noise-family composition of the clean C_2 preferred region.

Noise family	C_2-preferred cells
Bit flip	2
Dephasing	16
Depolarizing	6
Mixed Pauli	9
Phase flip	10

Table S3: Partial-syndrome preferred-policy counts by observed syndrome ratio.

r_{obs}	Cells	C_1	C_2	C_3	C_3^{R}
1.00	72	37	35	0	0
0.75	72	72	0	0	0
0.50	72	12	0	0	60
0.25	72	14	0	0	58

Table S4: Noisy-only preferred-policy counts by syndrome-bit noise probability. The robust policy remains inactive in this negative-control map.

p_{noise}	Cells	C_1	C_2	C_3	C_3^{R}
0.00	72	37	35	0	0
0.01	72	49	23	0	0
0.03	72	51	21	0	0
0.05	72	51	21	0	0
0.10	72	61	11	0	0

Table S5: Representative partial-plus-noisy policy-map rows showing regime cells where C_3^{R} is the preferred policy. Each row contains 72 cells.

r_{obs}	p_{noise}	C_1	C_2	C_3	C_3^{R}
0.50	0.03	13	0	0	59
0.50	0.05	12	0	0	60
0.50	0.10	15	0	0	57

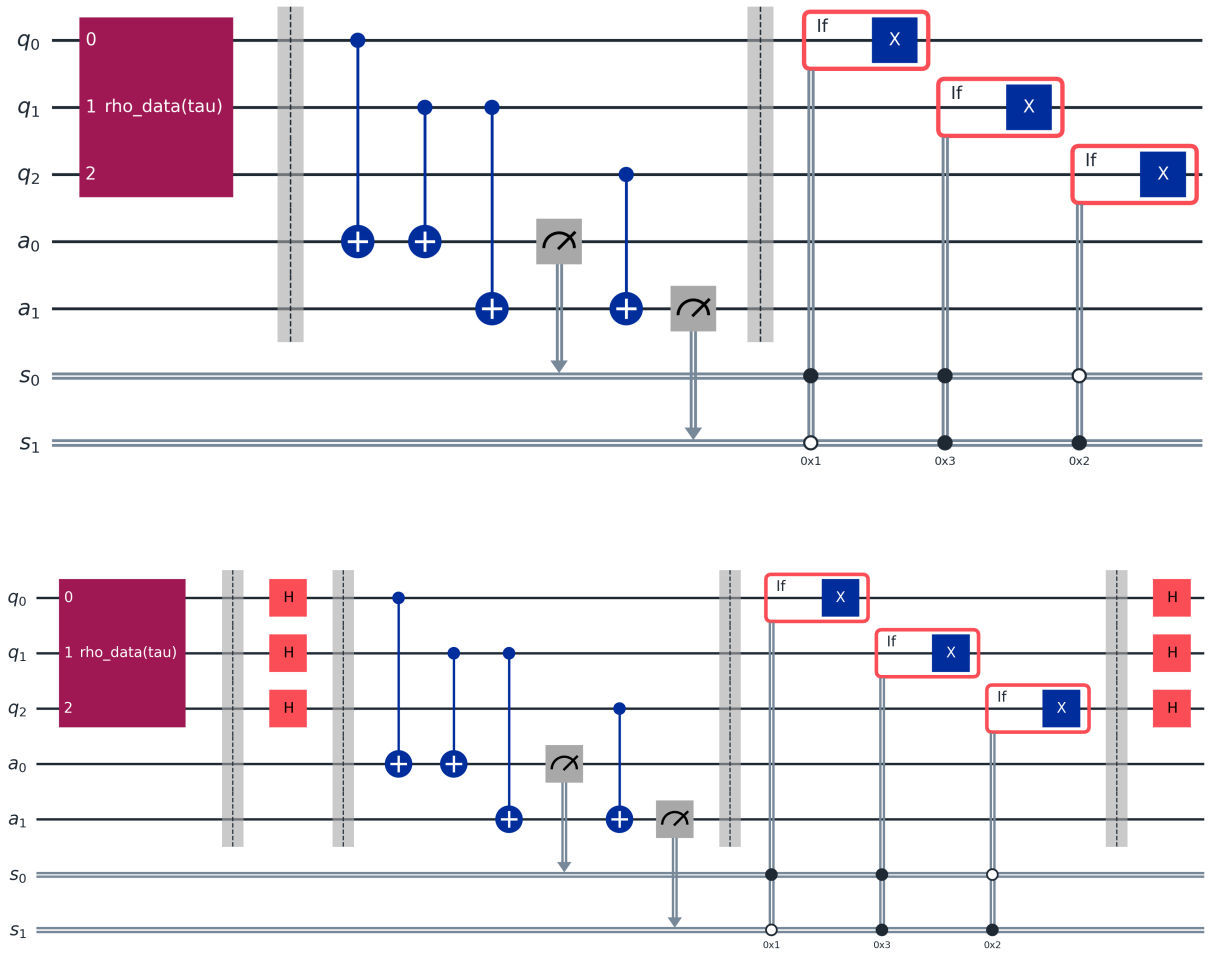


Figure S2: Syndrome-extraction and feed-forward recovery schematics. The upper panel shows the bit-flip recovery circuit and the lower panel shows the phase-flip recovery circuit. The state $\rho_{\text{data}}(\tau)$ denotes the clock-conditioned three-qubit data-register state injected before syndrome extraction. Ancillas a_0, a_1 store the two parity checks and are measured into the classical register s_0, s_1, with s_0 the least significant bit. The feed-forward integer convention is therefore $s_1 s_0$: 0x1 = 01 applies X to q_0, 0x3 = 11 applies X to q_1, and 0x2 = 10 applies X to q_2. For the phase-flip circuit, Hadamards before and after syndrome extraction move the same correction logic into the phase-flip protection basis. The simulations trace out the ancilla register after recovery; the schematic is drawn in system-qubit order q_0, q_1, q_2.

S7 Metric definitions

Let ΔF_C be the logical-fidelity gain of policy C. The false-safe and success thresholds are fixed before policy comparison:

$$\varepsilon_F = 0.01, \qquad F_{\text{success}} = 0.99. \tag{S7.1}$$

Central performance is summarized by mean gain, logical-success rate

$$P[F_C \geq F_{\text{success}}], \tag{S7.2}$$

and non-worsening rate

$$P[\Delta F_C \geq 0]. \tag{S7.3}$$

Downside risk is summarized by false-safe fidelity rate,

$$\text{FS}_F(C) = P[\Delta F_C < -\varepsilon_F], \tag{S7.4}$$

the lower quantile $q_{0.05}(C)$, and

$$\text{CVaR}_{0.05}(C) = \text{E}_{\text{emp}}[\Delta F_C \mid \Delta F_C \leq q_{0.05}(C)]. \tag{S7.5}$$

For C_3^{R}, the intervention metrics are conditional on C_2 proposing B. The block indicator is

$$B_R = \mathbf{1}[C_2 = B]\mathbf{1}[C_3^{\text{R}} = A]. \tag{S7.6}$$

Harmful and beneficial proposed switches are

$$H_2 = \mathbf{1}[C_2 = B]\mathbf{1}[\Delta F_A > \Delta F_B + \varepsilon_F], \tag{S7.7}$$

$$G_2 = \mathbf{1}[C_2 = B]\mathbf{1}[\Delta F_B > \Delta F_A + \varepsilon_F]. \tag{S7.8}$$

We report block rate, harmful-block precision, harmful-switch recall, beneficial-switch block rate, prevented loss, missed gain, and net intervention effect. Beneficial-switch retention is one minus the beneficial-switch block rate. The uncertainty-bin analysis groups rows by u_{raw} to check whether the robust policy behaves as an uncertainty-conditioned intervention rather than as an unconditional score adjustment. The scalar net intervention effect is

$$E_{\text{net}} = \sum_i B_{R,i}(\Delta F_{A,i} - \Delta F_{B,i}), \tag{S7.9}$$

equivalently,

$$E_{\text{net}} = \sum_i B_{R,i} \max\{0, \Delta F_{A,i} - \Delta F_{B,i}\} - \sum_i B_{R,i} \max\{0, \Delta F_{B,i} - \Delta F_{A,i}\}. \tag{S7.10}$$

Thus the same scalar can be read as prevented harmful-switch loss minus missed beneficial-switch gain. A positive value means that blocked switches improved aggregate fidelity relative to accepting the C_2 proposal. We also report $E_{\text{net}}/\sum_i B_{R,i}$ and $E_{\text{net}}/\sum_i \mathbf{1}[C_{2,i} = B]$ to separate aggregate benefit from the number of vetoed proposals. For offline comparison we also compute oracle regret,

$$\Delta F^\star = \max\{\Delta F_A, \Delta F_B\}, \qquad \text{Reg}(C) = \Delta F^\star - \Delta F_C. \tag{S7.11}$$

The oracle is not available to the policy; it is used only to quantify how far the chosen candidate is from the best candidate available on that simulation row. This diagnostic is especially useful when mean gain and false-safe rate point in different directions.

S8 Threshold and uncertainty diagnostics

This section audits the two threshold choices that could otherwise appear arbitrary: the uncertainty gate $u_{\max}$ in the robust policy and the admissibility calibration factor κ. The purpose is not to tune an optimum threshold after seeing the outcomes, but to show how the reported operating point behaves relative to nearby counterfactual settings and negative controls.

The default robust gate uses $u_{\max} = 0.99$. This value should not be interpreted as a calibrated posterior confidence level. It is a saturation guard for the additive stress index u_{raw} after clipping. Table S6 reports offline counterfactuals for the same rows under alternative $u_{\max}$ values.

Table S6: C_3^{R} uncertainty-threshold sensitivity. Mean gain and false-safe fidelity rate are recomputed from the same result rows after changing only the $u_{\max}$ gate. Noisy-only is included as a negative control; $u_{\max}$ has no effect there because no row triggers the robust gate.

Regime	$u_{\max}$	mean gain	FS_F	blocked	precision	ben. block	net/block
Noisy-only	any	0.1169	0.0678	0	–	–	–
Partial	0.75	0.1189	0.0340	543	0.8508	0.4833	0.0106
Partial	0.99	0.1189	0.0389	476	0.8697	0.3750	0.0119
Partial	1.00	0.1169	0.0694	0	–	–	–
Partial+noisy	0.75	0.1205	0.0000	821	0.8465	1.0000	0.0092
Partial+noisy	0.99	0.1212	0.0134	670	0.8791	0.6739	0.0136
Partial+noisy	1.00	0.1170	0.0685	0	–	–	–
Ambig.+meas.	0.75	0.1608	0.0000	480	0.7896	0.5316	0.0061
Ambig.+meas.	0.99	0.1604	0.0000	357	0.7759	0.4211	0.0044
Ambig.+meas.	1.00	0.1600	0.0000	0	–	–	–

For historical implementation documentation, the admissibility thresholds contain the calibration constant κ in $\theta_k = \mu_k + \kappa\sigma_k$. To check that the default $\kappa = 1.5$ is not a single-point artifact, we ran the recovery-ablation sweep with $\kappa \in \{1.0, 1.5, 2.0\}$ while holding the noise families, noise strengths, and reference-bank widths fixed. The sweep covers bit-flip, phase-flip, and dephasing stress scenarios at five strengths and three bank widths, giving 45 cases per calibration point.

Table S7: Historical calibration-sweep sensitivity for the submitted recovery objective. Recovered admissible is the calibrated-admissibility rate of the final recovered trajectory. Candidate improvement is the rate at which the optional Stage-2 admissible candidate improves the recovery objective relative to Stage 1, and helpful tradeoff additionally requires non-worse clean distance.

κ	cases	rec. admiss.	cand. improve	mean cand. ΔJ	helpful
1.0	45	0.2000	0.5556	-0.1100	0.1111
1.5	45	1.0000	0.6667	0.1702	0.1111
2.0	45	1.0000	0.6667	0.1702	0.1111

The default $\kappa = 1.5$ is on the same plateau as $\kappa = 2.0$ in this sweep, whereas the stricter $\kappa = 1.0$ calibration sharply reduces recovered admissibility and gives a negative mean candidate objective gain. The helpful tradeoff rate is invariant across κ in this sweep, indicating that these κ changes affect admissibility coverage and Stage-2 candidate acceptance more than the underlying clean-distance behavior. This supports using $\kappa = 1.5$ as a historical implementation tolerance, but it does not identify a discriminative admissibility gate on the submitted main grid: candidate B is admissible in every row. No Stage-2 refinements were accepted by the conservative application rule in this sweep, so Table S7 is retained as calibration documentation rather than policy-performance evidence.

Table S8: Uncertainty-bin intervention summary at the default $u_{\max} = 0.99$. Bins with zero blocks show that the gate is concentrated near saturated syndrome stress rather than acting as an unconditional score penalty.

Regime	u_{raw} bin	cases	$C_2 = B$	blocked	block rate	precision	net
Partial	< 1.00	1638	618	0	0.0000	–	0.0000
Partial	$[1, \infty)$	1242	476	476	1.0000	0.8697	5.6616
Partial+noisy	$[0.75, 1.00)$	516	180	29	0.1611	0.6897	-0.3097
Partial+noisy	$[1, \infty)$	1644	641	641	1.0000	0.8877	9.4309
Ambig.+meas.	< 1.00	2142	538	0	0.0000	–	0.0000
Ambig.+meas.	$[1, \infty)$	1458	357	357	1.0000	0.7759	1.5749

These diagnostics show that the default $u_{\max} = 0.99$ gate behaves primarily as a saturation guard. When $u_{\max} = 1.00$, the veto is disabled in the reported degraded regimes and C_3^{R} collapses to C_2. When $u_{\max} = 0.75$, the controller becomes more conservative, reducing false-safe failures further in some regimes but increasing beneficial-switch loss. The negative net value in the partial-plus-noisy $[0.75, 1.00)$ bin is a small borderline-stress region where the default gate vetoes some beneficial C_2 switches; the aggregate partial-plus-noisy benefit is dominated by the saturated-stress bin $[1, \infty)$. In the $u_{\max} = 0.75$ counterfactual for the same regime, the aggressive threshold blocks all beneficial C_2 switches, eliminating the upside from the relational candidate. Therefore, the default operating point should be interpreted as a safety-biased but not optimized threshold.

S9 Supplementary result summaries

This section reports the numerical values underlying the main Results. The main text emphasizes operating-zone interpretation and downside-risk behavior; the tables below retain additional central-performance, oracle, and intervention diagnostics.

Table S9: Historical submitted-grid comparison between C_2 and C_3^{R}. Entries are reported as C_2/C_3^{R} and are retained for audit only.

Regime	n	mean gain	logical success	non-worsen	FS_F
Clean	2400	0.1273/0.1273	0.6358/0.6358	0.9554/0.9554	0.0238/0.0238
Partial	2880	0.1169/0.1189	0.4552/0.5976	0.8774/0.9306	0.0694/0.0389
Noisy-only	3600	0.1169/0.1169	0.4556/0.4556	0.8814/0.8814	0.0678/0.0678
Partial+noisy	2160	0.1170/0.1212	0.4551/0.7241	0.8792/0.9755	0.0685/0.0134
Ambig.+meas.	3600	0.1600/0.1604	0.5208/0.5978	1.0000/1.0000	0.0000/0.0000

Table S10: Lower-tail and oracle-regret diagnostics. Entries are reported as C_2/C_3^{R}.

Regime	$q_{0.05}$	$\text{CVaR}_{0.05}$	mean oracle regret	raw uncertainty mean
Clean	0.0000/0.0000	-0.0033/-0.0033	0.0132/0.0132	0.0000
Partial	-0.0152/-0.0006	-0.0208/-0.0147	0.0271/0.0251	0.7604
Noisy-only	-0.0104/-0.0104	-0.0207/-0.0207	0.0270/0.0270	0.0380
Partial+noisy	-0.0104/0.0000	-0.0206/-0.0034	0.0271/0.0228	1.0989
Ambig.+meas.	0.0192/0.0238	0.0078/0.0084	0.0336/0.0332	0.8261

The intervention table should be read together with the negative controls. In the clean and noisy-only regimes, C_3^{R} blocked no C_2 switches. Thus the non-zero intervention rows are not evidence of a global conservative bias; they identify the regimes where the uncertainty and admissibility gates made the C_2 proposal unsafe enough to veto.

Table S11: C_3^{R} intervention quality in regimes with non-zero blocking.

Regime	$C_2 = B$	blocked	block rate	precision	recall	ben. block	net effect
Partial	1094	476	0.4351	0.8697	0.4461	0.3750	5.6616
Partial+noisy	821	670	0.8161	0.8791	0.8475	0.6739	9.1212
Ambig.+meas.	895	357	0.3989	0.7759	0.3929	0.4211	1.5749

Table S12: Normalized intervention effects. Prevented and missed values are aggregate fidelity-gain differences over blocked C_2 switches.

Regime	blocked	prevented	missed	net	net/block	net/$(C_2 = B)$
Partial	476	12.2670	6.6055	5.6616	0.0119	0.0052
Partial+noisy	670	17.4350	8.3139	9.1212	0.0136	0.0111
Ambig.+meas.	357	8.3526	6.7777	1.5749	0.0044	0.0018

S10 Revision audit and new experiments

S10.1 Status of the submitted grid

The submitted grid is retained as an immutable audit dataset, but it is not used as efficacy evidence in the revision. Its encoded system Hamiltonian was zero while the clock was initialized as $|+\rangle_C$ under $H_C = 0.8X_C$. The conditional-slice audit therefore finds constant normalized system densities (maximum change 3.51×10^{-16}) and a nonzero global-constraint residual (0.8). In particular, the old clock-orbit labels differ only by a global phase for this initialization, so their projectors do not provide nine independent observation windows. This also resolves the apparent endpoint-overlap issue: the overlap formula is a property of the submitted orbit, not evidence of a nondegenerate conditional trajectory.

For clarity, the submitted encoded preparation was

$$|\Psi_{\mathrm{sub}}\rangle = c_0|0\rangle_C|0_L\rangle + c_1|1\rangle_C|1_L\rangle, \qquad H_S = 0, \tag{S10.1}$$

with the clock initialized in $|+\rangle_C$. This explicit construction is reported because the physical qualification of a nontrivial conditional trajectory was absent from the submitted experiment.

The counterfactual $u_{\max} = 1.00$ removes the uncertainty conjunct. On the submitted grid the remaining score/admissibility conditions block zero proposed switches in every active regime. We consequently interpret the submitted downside-risk effect as uncertainty-threshold control, not an independently identified relational-spectral effect.

Candidate B was marked admissible in every submitted row (2,400 clean, 2,880 partial, 3,600 noisy-only, 2,160 partial-plus-noisy, and 3,600 ambiguity-plus-measurement rows). Thus the submitted admissibility condition did not supply an independent veto. The seed-wise historical differences are shown in Table S13; these intervals are conditional on the fixed grid and ten seed blocks, and are not out-of-grid or confirmatory inference. The row-level blocked-switch differences and the source manifest are released with the audit artifacts.

For complete reproducibility of that historical implementation, the recovery weights were

$$(\alpha, \beta, \lambda_1, \lambda_2, \lambda_3) = (64.0, 16.0, 0.75, 0.10, 0.10), \tag{S10.2}$$

with an additional reference penalty weight zero. The conservative policy-facing syndrome support for candidate A was

$$s_A = \frac{1}{m}\sum_{r=1}^{m} p_r(00)\, I(o_r)\, c(o_r), \qquad s_B = 0, \tag{S10.3}$$

where $p_r(00)$ is the simulated no-error syndrome probability, $I(o_r)$ is the observed-information fraction (1 for two known bits, 1/2 for an ambiguous symbol, and the fraction of known bits otherwise), and $c(o_r)$ is 1 when the observed symbol is compatible with 00, 1/2 for the equal-ambiguity symbol, and 0 otherwise. Candidate A itself was the complete circuit syndrome-recovery channel applied to the underlying noisy density at every slice; missing, corrupted, and ambiguous syndrome symbols affected only this controller support and its uncertainty proxy. It was therefore not a decoder operating solely on the damaged observed syndrome. This limitation is explicit in the revision.

The differing submitted sample sizes were caused by different numbers of swept parameter combinations, each with ten seeds: clean 240 combinations (2,400 rows), partial 288 (2,880), noisy-only 360 (3,600), partial-plus-noisy 216 (2,160), and ambiguity-plus-measurement 360 (3,600). The column formerly named "true failure boundary" is removed because it was not a prespecified metric. The standalone clean map and the zero-stress endpoints reported elsewhere in the submitted material are distinct parameter sweeps, not duplicate representations of the same rows; neither is used as revised efficacy evidence.

Table S13: Historical submitted-grid paired differences, $C_3^{\mathrm{R}} - C_2$. False-safe differences are percentage points; negative is favorable. Intervals are percentile bootstrap intervals over the ten fixed seed blocks and are reported for audit, not as confirmation of the withdrawn efficacy claim.

Regime	mean-gain difference [95% CI]	false-safe difference [95% CI]
Clean	0.00000 [0.00000, 0.00000]	0.000 [0.000, 0.000]
Partial	0.00197 [0.00053, 0.00347]	−3.056 [−3.472, −2.639]
Noisy-only	0.00000 [0.00000, 0.00000]	0.000 [0.000, 0.000]
Partial+noisy	0.00422 [0.00122, 0.00717]	−5.509 [−6.574, −4.398]
Ambig.+meas.	0.00044 [−0.00097, 0.00186]	0.000 [0.000, 0.000]

Table S14: Distribution of row-level historical blocked-switch effects in the three submitted regimes with nonzero interventions. The effect is $d = \Delta F_A - \Delta F_B$. Consistent with Table S11, a blocked proposed switch is harmful when $d > \varepsilon_F$, beneficial when $d < -\varepsilon_F$, and a near tie otherwise, where $\varepsilon_F = 0.01$. Quantiles use all blocked rows; these are descriptive audit summaries, not efficacy evidence.

Regime	rows	harmful	near tie	beneficial	$q_{0.05}/q_{0.50}/q_{0.95}$
Partial	476	414	17	45	−0.14985/0.03015/0.03015
Partial+noisy	670	589	19	62	−0.06485/0.03015/0.03015
Ambig.+meas.	357	277	0	80	−0.14985/0.03015/0.03015

S10.2 R1: energy-matched finite-clock qualification

The revision construction uses $H_C = \omega Z_C$ and $H_S = -\omega(n_x X_L + n_z Z_L)$ with $\omega = 0.8$. For an arbitrary logical state $|\psi\rangle = c_+|n_+\rangle + c_-|n_-\rangle$, the prepared joint state is

$$|\Psi\rangle = c_+|0\rangle_C|n_+\rangle_L + c_-|1\rangle_C|n_-\rangle_L. \tag{S10.4}$$

It obeys $(H_C + H_S)|\Psi\rangle = 0$. We use the eight effects

$$E_k = \frac{2}{8}|\tau_k\rangle\langle\tau_k|, \qquad \tau_k = \frac{k\pi}{8\omega}, \qquad k = 0, \ldots, 7, \tag{S10.5}$$

which sum to the identity but are not orthogonal projectors. The tilted setting $n = (1, 0, 1)/\sqrt{2}$ has rank-three population-observation design, whereas the aligned setting has rank one. Across the qualified R1 construction, the maximum constraint residual is 1.58×10^{-16}, the maximum conditional-density gap is 1.01×10^{-15}, the POVM-completeness error is 1.26×10^{-16}, and the minimum adjacent-label geometry is 0.14645. R1 is a multi-copy observability qualification only; it is not an unknown-state QEC or recovery-efficacy experiment.

S10.3 R2: superseded preliminary gate-definition run

R2 is retained as an archival preliminary analysis, not as final efficacy evidence. Although it used disjoint reference, calibration, and target streams, its reported validation comparison refitted the continuous candidate A to the validation labels. It therefore compared a fixed template candidate with a validation-refitted reference rather than scoring two unchanged fitting-stage candidates. Its intervals also resampled target pairs only after measurement seed averaging. R2 is consequently superseded by the corrected R3 design below; its protocol, code, raw results, and manifest remain released for audit.

S10.4 R3: corrected held-out template-switching risk test

R3 tests whether a held-out residual can reject harmful switches from a continuous count-record estimate to a direct nearest-template baseline. A 32-state template bank, 32 calibration targets, and 64 test targets arranged in 32 antipodal pairs are disjoint within each stream. The switching policy uses only fitting and validation counts, the known Hamiltonian, and the fixed bank; the target density is used only for scoring after the decision. The direct observation-space candidate B_{fit} is the nearest template under the same fitting-label loss as A_{fit}. An oracle target-fidelity comparison is used offline only as a diagnostic.

For each target and measurement seed, four of eight labels are fitting labels and the complementary four are validation labels. The continuous candidate A_{fit} and B_{fit} are both fit once using fitting labels, then kept unchanged. With the weighted count loss L_{val} on the validation labels, the held-out residual is

$$h = L_{\mathrm{val}}(B_{\mathrm{fit}}) - L_{\mathrm{val}}(A_{\mathrm{fit}}). \tag{S10.6}$$

No candidate is refitted on validation labels. The policies are continuous A, direct template B, score-only C_2, uncertainty-only U, residual-only H, and their combined gate C_3^{R}. In each stream, the score, uncertainty, and held-out residual thresholds are set once to the 75th empirical quantiles of that stream's separate calibration set at the fixed bit-flip, 1,024-copy, physical-noise 0.05, readout-flip 0.05 operating point. Their values may differ between streams, but they remain fixed across all 24 held-out test cells within each stream. No test target is used to set or retune them.

The grid has two coupled basis-equivalent code conventions, two copy budgets (256 and 1,024), three physical-noise probabilities (0.01, 0.05, 0.10), two readout-flip probabilities (0, 0.05), 12 measurement seeds, and 32 target pairs: 24 cells per stream. R3 was run on four predefined data streams. Stream r0 reuses R2's random streams solely to isolate the corrected gate definition; r1–r3 use independent bank, calibration, target, and measurement streams. The two code conventions are checks, not independent replications. Reported 95% intervals are paired two-way cluster bootstrap intervals resampling both target pairs and measurement seeds; calibration-threshold uncertainty is not included.

In every one of the 96 evaluated cell–stream combinations, $H - C_2$ has a negative false-safe-rate interval excluding zero. This is an unconditional rate comparison between nested acceptance sets; a lower rate does not by itself establish selective rejection of worse proposals. R4 below

Table S15: R3 grid summary. A negative false-safe interval means fewer harmful template switches. A positive $C_3^{\mathrm{R}} - A$ estimate denotes higher mean target fidelity, but its intervals are not uniformly positive.

Stream	$H - C_2 < 0$	$C_3^{\mathrm{R}} - U < 0$	$C_3^{\mathrm{R}} - A > 0$	Grid mean $C_3^{\mathrm{R}} - A$
r0	24/24	12/24	0/24	−0.001184
r1	24/24	8/24	2/24	−0.000550
r2	24/24	12/24	12/24	+0.001251
r3	24/24	6/24	6/24	−0.000011

tests that question at equal expected switching rates. The additional conjunction is more limited: $C_3^{\mathrm{R}} - U$ is negative in 12/24, 8/24, 12/24, and 6/24 cells in r0–r3, respectively, and the mean-fidelity comparison $C_3^{\mathrm{R}} - A$ has no consistent sign.

At the middle operating point (bit-flip, 1,024 copies, physical noise 0.05, readout flip 0.05), Table S16 gives the two-way intervals. The result records the reduction of harmful direct-template switches in this simulator. Its interpretation requires the equal-rate and continuous-estimator comparisons in R4; the unconditional contrast alone does not establish an advantage in decision quality, continuous estimation, or spectral representation.

Table S16: R3 middle-point paired differences. False-safe differences are percentage points; negative is favorable. Fidelity differences are evaluator-only target metrics.

Stream	$H - C_2$ FS [95% CI]	$C_3^{\mathrm{R}} - U$ FS [95% CI]	$C_3^{\mathrm{R}} - A$ ΔF [95% CI]
r0	−14.844 [−21.615, −8.854]	−7.813 [−13.281, −3.646]	−0.002308 [−0.006547, 0.002417]
r1	−8.333 [−13.281, −4.167]	−3.125 [−6.250, −0.781]	−0.001203 [−0.005841, 0.003377]
r2	−10.677 [−17.187, −5.208]	−6.771 [−11.979, −2.604]	0.002559 [−0.000496, 0.005887]
r3	−5.990 [−10.417, −2.604]	−1.823 [−4.427, −0.254]	0.003306 [−0.002715, 0.009931]

S11 Switching-rate and continuous-estimator audit (R4)

R4 is a reviewer-prompted secondary analysis of the four existing R3 streams. The analysis rules were fixed on 29 October 2026, after the R3 results had been inspected and before the new contrasts were calculated. This is not a new independent replication or an externally registered analysis. No new target or measurement realization is introduced. We reconstructed the observations from their archived seeds and verified all 73,728 record diagnostics, fidelities, count totals, fitting masks, and gate decisions against the saved R3 records. The maximum discrepancy in the checked numerical fields was zero.

S11.1 Physical role of the finite clock

For the R3 readout, the data likelihood is

$$p(k, z \mid \boldsymbol{r}) = \frac{1}{16}\left[1 + z(1-2\eta)\boldsymbol{d}_k^{\mathsf{T}}\boldsymbol{r}\right]. \tag{S11.1}$$

Here k is one of eight labels, $z = \pm 1$ is the logical-population outcome, η is the known readout-flip probability, and the known design row $\boldsymbol{d}_k$ includes rotation and code-matched physical noise. Conditioning the energy-matched stationary joint state realizes this observation model. Uniformly choosing eight ordinary population measurement settings realizes the same model. Therefore the clock supplies a consistent physical interpretation, but is not required by the switching statistic. No

graph-spectral feature enters R3 or R4, and no performance effect is assigned to the Page–Wootters formalism itself.

S11.2 Original gates at a common expected switching rate

Let S_g be the accepted proposals of original fixed-threshold gate g within one cell and stream. Since $S_{C_3^{\mathrm{R}}} \subseteq S_U \cap S_H \subseteq S_{C_2}$, set $m = |S_{C_3^{\mathrm{R}}}|$ and thin each $g \in \{C_2, U, H\}$ uniformly with probability $m/|S_g|$. The per-record switching probability is

$$\pi_{ig} = \mathbf{1}\{i \in S_g\}\frac{m}{|S_g|}, \qquad \sum_i \pi_{ig} = m. \tag{S11.2}$$

An empty acceptance set has zero switching probability. We integrate this randomization analytically. The C_2 version is a random-retention control, denoted R. This operation preserves the relative selection within each original gate; it does not tune a threshold using target fidelity.

Define $b_i = \mathbf{1}\{F_{Bi} < F_{Ai} - 10^{-12}\}$. The conditional harmful fraction is $\sum_i \pi_{ig} b_i / \sum_i \pi_{ig}$, and total mean fidelity is $N^{-1}\sum_i [F_{Ai} + \pi_{ig}(F_{Bi} - F_{Ai})]$, with $N = 768$. When $m = 0$, conditional harm is undefined and the cell cannot support a selection-quality contrast. This occurs in all twelve 256-copy cells of each stream. All twelve 1,024-copy cells have positive common coverage. Table S17 reports both conditional quality and net fidelity at this common expected coverage.

S11.3 Fixed-retention batch rankings

Within the original C_2 proposal set, we compare retention fractions $q \in \{0.25, 0.50, 0.75\}$. Proposals are ranked by fitting score, uncertainty u, residual h, or the combined statistic

$$c = \max\left(\frac{u}{t_u}, \frac{h}{t_h}\right), \tag{S11.3}$$

where $t_u, t_h > 0$ are the original stream-specific R3 calibration thresholds. Accepting $c \le a$ rescales the original two thresholds by the same factor a and recovers their conjunction at $a = 1$. These ranking families are distinct from the original fixed-threshold gates.

Lower scores are accepted first. Scores are rounded to twelve decimal places only for numerical tie handling. If a cutoff falls inside a tie group, all group members receive the same switching probability so that exactly $q|S_{C_2}|$ proposals are accepted in expectation. Random retention accepts every proposal with probability q. All five rules therefore have identical expected switching rates within each cell, not merely similar grid averages. Cutoffs depend on the observed batch diagnostics without target outcomes. They characterize batch selection and are not calibration-only deployment thresholds. All three prespecified fractions are reported in Table S18; no best fraction is selected from the test fidelities.

S11.4 Optimized continuous estimators and statistics

The original radial projection A is feasible but need not minimize weighted least squares over the Bloch ball. We add global constrained minimizers A^*_{fit} and A^*_{all} using fitting labels or all labels, respectively. For Gram matrix G and right-hand side $\boldsymbol{b}$, the solution is $\boldsymbol{x} = (G + \lambda I)^{-1}\boldsymbol{b}$, with $\lambda = 0$ when the unconstrained solution is inside the ball; otherwise $\lambda > 0$ is chosen so that $\|\boldsymbol{x}\| = 1$. A pseudoinverse handles singular unconstrained systems. The objective is convex. Independent constrained-solver tests, feasibility checks, KKT stationarity checks, and comparison with the original A loss verify the implementation. Across all replayed fits, the maximum normalized

KKT residual was 6.13×10^{-15}, the maximum norm was $1 + 5.62 \times 10^{-14}$, and the maximum numerical loss excess over A was 3.47×10^{-17}.

The full-record control uses the same prepared-copy budget as the residual gate, which already reads the validation labels. Its advantage therefore measures a relevant opportunity cost of the switching task. The bank is not assumed to be a necessary output alphabet in an application, and these controls minimize the declared count loss rather than oracle target fidelity. Table S19 shows why a selection-quality benefit does not imply an overall advantage over full-record continuous estimation.

We use 5,000 paired two-way cluster bootstrap samples per stream, resampling the twelve measurement seeds and thirty-two target pairs while retaining their coupled antipodal signs. A shared resample across cells preserves their common seed/pair axes in stream averages. Each resample recalculates thinning factors and ranking cutoffs. Intervals are pointwise 95% intervals, conditional on the bank, model, and R3 calibration thresholds; there is no family-wise significance claim. The coupled code conventions are not independent replications. Stream means equally weight the 24 cells; conditional-harm means for the original common-coverage comparison use only the twelve positive-coverage cells.

Complete cell summaries also report unconditional harmful-switch rate, expected switching rate, mean fidelity, and descriptive per-record $q_{0.05}/\mathrm{CVaR}_{0.05}$. For these tails, each record contributes mass $1 - \pi_i$ at F_{Ai} and mass π_i at F_{Bi}; the lower 5% mass is integrated exactly, including a fractional boundary. This unit differs from R3's seed-averaged target-pair tail summaries. The archived R3 JSON field named `cvar05_exact_mass` averages the two lowest of thirty-two pair means, representing 6.25% mass rather than an exact fractional 5%. This archival field is not used in the main-text selection or mean-fidelity contrasts; R4 uses the exact 5% mixture definition above. Protocol, analysis source, all inputs, replay checks, and complete outputs accompany the revision in the reproducibility archive.

Table S17: Original fixed gates after analytic random thinning to the common expected switching rate of C_3^{R}. Conditional-harm differences are percentage points and are equally averaged over the 12 positive-coverage cells (all 1,024-copy cells). Mean-fidelity differences average all 24 cells, including zero interventions at 256 copies. Brackets are pointwise paired two-way 95% intervals.

Stream	Active	$H - R$: harm	$C_3^{\mathrm{R}} - U$: harm	$C_3^{\mathrm{R}} - U$: ΔF
r0	12/24	-7.62 $[-10.26, -5.42]$	-5.23 $[-7.56, -3.32]$	$+0.00178$ $[+0.00134, +0.00214]$
r1	12/24	-3.00 $[-4.36, -1.81]$	-1.46 $[-2.39, -0.69]$	$+0.00065$ $[+0.00037, +0.00094]$
r2	12/24	-6.19 $[-9.48, -3.88]$	-4.37 $[-7.05, -2.45]$	$+0.00111$ $[+0.00076, +0.00140]$
r3	12/24	-4.48 $[-7.04, -2.61]$	-1.93 $[-3.26, -0.98]$	$+0.00085$ $[+0.00047, +0.00119]$

Table S18: Equal-retention batch rankings. Retention q is the percentage of original C_2 proposals accepted in expectation. R denotes uniform random retention. Harm is the percentage of accepted switches with $F_B < F_A - 10^{-12}$, equally averaged across all 24 cells per stream. All rankings have the same expected switching rate in each cell and bootstrap sample; cutoffs use observed diagnostics without target outcomes. The last column is a paired pointwise 95% interval in percentage points.

Stream	q	R harm	U harm	H harm	Combined harm	$H - U$: harm
r0	25	65.55	66.00	28.20	62.18	−37.80 [−42.81, −30.29]
r0	50	65.55	66.38	45.61	63.10	−20.78 [−24.88, −16.24]
r0	75	65.55	66.37	56.45	63.36	−9.93 [−11.99, −7.72]
r1	25	56.88	59.18	20.26	58.17	−38.92 [−42.77, −32.01]
r1	50	56.88	58.43	36.67	57.55	−21.76 [−25.48, −17.00]
r1	75	56.88	57.76	47.61	56.62	−10.16 [−12.04, −7.94]
r2	25	51.75	51.73	15.00	48.50	−36.73 [−41.18, −31.50]
r2	50	51.75	51.54	27.86	48.13	−23.68 [−26.55, −19.41]
r2	75	51.75	51.02	40.26	47.20	−10.76 [−12.77, −8.21]
r3	25	53.97	55.18	18.05	53.85	−37.13 [−41.69, −30.94]
r3	50	53.97	54.25	30.75	53.05	−23.50 [−26.59, −19.14]
r3	75	53.97	53.60	42.13	52.14	−11.47 [−13.46, −8.89]

Table S19: Continuous controls and the 50%-retention H ranking, averaged equally across the 24 cells. A is the original radial projection; A^*_{fit} globally minimizes the fitting-label weighted loss over the Bloch ball; A^*_{all} uses the entire prepared-copy budget. The two right columns are paired H-minus-control mean-fidelity differences with pointwise 95% intervals. The optimized-loss estimators are not oracle fidelity optimizers.

Stream	F_A	$F_{A^*_{\text{fit}}}$	$F_{A^*_{\text{all}}}$	H minus A^*_{fit}	H minus A^*_{all}
r0	0.962411	0.964404	0.981577	+0.00522 [+0.00114, +0.00872]	−0.01195 [−0.01600, −0.00846]
r1	0.962731	0.964732	0.981855	+0.00815 [+0.00468, +0.01145]	−0.00897 [−0.01245, −0.00580]
r2	0.963577	0.965622	0.982381	+0.00676 [+0.00360, +0.00971]	−0.01000 [−0.01336, −0.00707]
r3	0.961308	0.963219	0.981184	+0.00847 [+0.00478, +0.01161]	−0.00949 [−0.01350, −0.00611]

S11.5 Harm-margin sensitivity and direct H–U fidelity comparison

The primary R4 label uses $F_B < F_A - 10^{-12}$, where 10^{-12} is a numerical tolerance rather than a practical loss threshold. Following further feedback, a separate follow-up analysis applies all margins $\varepsilon \in \{10^{-12}, 0.001, 0.005, 0.01\}$ under

$$b_i(\varepsilon) = \mathbf{1}\{F_{Bi} < F_{Ai} - \varepsilon\}. \tag{S11.4}$$

The positive margins are in absolute fidelity units. The value 0.01 matches the size of the historical audit tolerance, but the historical and R3/R4 constructions and target metrics remain distinct. No application-specific utility threshold is inferred from this choice.

All candidate states, proposals, calibration thresholds, ranking scores, retention levels, tie rules, and matching factors are unchanged. Only the evaluator's harm label changes. The analysis retains every stream, cell, and declared margin, and reuses the original 5,000 paired two-way bootstrap draws, recalculating batch cutoffs and thinning factors in each draw. The margin and reporting rules were fixed before computing the follow-up outcomes, after the original R4 results were known. This remains an additional secondary analysis, not a new independent replication. The original R3 and R4 inputs, protocols, and outputs are preserved. The reproduced original estimates and intervals agree to within 5.4×10^{-16}.

Table S20 reports the 50%-retention H–U conditional-harm comparison at every declared margin. At $\varepsilon = 0.01$, differences remain negative in all four stream averages, with paired pointwise 95%

intervals below zero. This shows that the observed contrast is not confined to numerically small fidelity losses.

Table S21 directly reports H minus U overall mean fidelity at each of the three original retention levels. The signed-fidelity calculation keeps every gain and loss and does not discard differences inside the harm margin. The comparison is paired at the same expected switching rate, with each stream equally averaging all 24 cells. These results concern batch rankings that use observed diagnostics without target outcomes; they do not establish a calibrated deployment policy or remove the full-record continuous-estimator limit. The accompanying reproducibility archive contains the separate follow-up protocol, source, and all-cell sensitivity summaries.

Table S20: Harm-margin sensitivity at 50% retention of the original C_2 proposals. A switch is harmful when $F_B < F_A - \varepsilon$. H and U are the unchanged batch-ranking rules, with the same expected switching rate in each cell. Harm columns are percentages among accepted switches, equally averaged over all 24 cells; the last column is H minus U in percentage points with a paired pointwise 95% interval. The positive margins are sensitivity choices in absolute fidelity units, not validated utility thresholds.

Stream	ε	H harm (%)	U harm (%)	$H-U$: harm [95% CI]
r0	10^{-12}	45.61	66.38	-20.78 $[-24.88, -16.24]$
r0	0.001	43.41	64.98	-21.57 $[-26.22, -16.56]$
r0	0.005	34.84	59.17	-24.33 $[-29.25, -18.79]$
r0	0.01	25.43	52.03	-26.60 $[-30.53, -21.41]$
r1	10^{-12}	36.67	58.43	-21.76 $[-25.48, -17.00]$
r1	0.001	33.40	55.98	-22.58 $[-26.48, -17.51]$
r1	0.005	25.12	49.61	-24.49 $[-28.14, -19.34]$
r1	0.01	16.89	40.96	-24.08 $[-27.20, -19.45]$
r2	10^{-12}	27.86	51.54	-23.68 $[-26.55, -19.41]$
r2	0.001	23.55	48.32	-24.77 $[-27.48, -20.34]$
r2	0.005	13.89	39.61	-25.72 $[-28.32, -20.98]$
r2	0.01	8.69	32.49	-23.80 $[-26.98, -18.55]$
r3	10^{-12}	30.75	54.25	-23.50 $[-26.59, -19.14]$
r3	0.001	26.94	51.25	-24.31 $[-27.34, -19.70]$
r3	0.005	12.80	39.13	-26.33 $[-29.58, -20.63]$
r3	0.01	7.47	32.71	-25.24 $[-29.71, -17.97]$

Table S21: Direct paired H-minus-U overall mean-fidelity comparison at equal expected switching rates. Retention q is relative to the original C_2 proposal set. Each stream equally averages all 24 cells. These are batch rankings using observed diagnostics without target outcomes, not calibration-only deployment policies. Signed fidelity differences retain every gain and loss and are unaffected by the harm margin. Brackets are paired pointwise two-way 95% intervals.

Stream	q (%)	F_H	F_U	$H-U$: ΔF [95% CI]
r0	25	0.969504	0.960704	$+0.00880$ $[+0.00728, +0.01027]$
r0	50	0.969628	0.958786	$+0.01084$ $[+0.00898, +0.01275]$
r0	75	0.965918	0.956227	$+0.00969$ $[+0.00791, +0.01129]$
r1	25	0.971629	0.962688	$+0.00894$ $[+0.00717, +0.01039]$
r1	50	0.972885	0.963116	$+0.00977$ $[+0.00811, +0.01141]$
r1	75	0.971967	0.963307	$+0.00866$ $[+0.00733, +0.00981]$
r2	25	0.970597	0.964283	$+0.00631$ $[+0.00492, +0.00770]$
r2	50	0.972381	0.964903	$+0.00748$ $[+0.00605, +0.00879]$
r2	75	0.972108	0.965642	$+0.00647$ $[+0.00517, +0.00758]$
r3	25	0.969498	0.961470	$+0.00803$ $[+0.00635, +0.00966]$
r3	50	0.971692	0.961545	$+0.01015$ $[+0.00815, +0.01197]$
r3	75	0.971051	0.961273	$+0.00978$ $[+0.00791, +0.01139]$